Online Appendix to: Frequent Losers: A Participation-Based Screen for Public Procurement

Darcio Genicolo-Martins[1,*], Paulo Furquim de Azevedo[1]

[1] *INSPER, São Paulo, Brazil*

Abstract

This online appendix accompanies *Frequent Losers: A Participation-Based Screen for Public Procurement* and contains: (A) the cover-bidding framework and proofs of the three formal results invoked in the main text; (B) core supporting tables and figures; (C) identification audits (Oster bounds and the anti-leakage audit); (D) threshold and specification robustness tables; (E) operational metrics and detector comparison; and (F) the staggered-design estimators that return null at the available bandwidths. References to sections, equations, and tables in the main text use the same labels as the published manuscript.

Appendix A. Cover-Bidding Framework and Proofs

This appendix formalizes the separating-equilibrium framework referenced throughout the main text. The framework supports four results that the empirical exercise reads against. A lemma establishes the wins = 0 filter as the equilibrium choice of the cover-bidder type (Lemma 1). A proposition identifies $\log(1 + \text{tenders\_count})$ as a sufficient ranking statistic for the cover-bidder type given award-record data (Proposition 2). A proposition characterizes the cartel's optimal deployment rate m^* in the principal cost of detection θ_k (Proposition 3). And a final proposition formalizes why the broad-sample association $\beta = \mathbb{E}[y \mid \text{losers} = 1, \mathbf{x}] - \mathbb{E}[y \mid \text{losers} = 0, \mathbf{x}]$ and the overlap-restricted association $\beta^{\text{ov}} = \mathbb{E}[y \mid \text{losers} = 1, \mathbf{x}, \text{overlap}] - \mathbb{E}[y \mid \text{losers} = 0, \mathbf{x}, \text{overlap}]$ are

*Corresponding author.
Email address: darciogm1@insper.edu.br (Darcio Genicolo-Martins)

different empirical objects under the framework's deployment problem, with the contrast between them diagnostic of the deployment sorting that the screening statistic captures (Proposition 4). The framework is an organizing device for the empirical patterns; it is not a source of independent causal identification.

Setup. A cartel of n firms in a procurement market controls a designated winner with bid b^* and deploys $m \geq 0$ cover bidders, each submitting a bid $b_\ell > b^*$. Per-bidder deployment cost is $c_1 > 0$; the cartel faces a detection probability $\theta_k \in (0, 1)$ that depends on oversight capacity in market k (smaller, less-supervised buyers have lower θ_k). The cartel maximizes expected surplus $\pi(m, \theta_k, c_1)$, the difference between cartel rents and the sum of cover-bidder costs and expected detection penalties. We impose two standard assumptions:

A1 (Profit positivity). $\partial\pi/\partial m\big|_{m=0} > 0$: the first cover bidder generates positive marginal cartel surplus, otherwise no cartel deploys cover.

A2 (Diminishing returns). $\partial^2\pi/\partial m^2 < 0$: the surplus function is strictly concave in m, so an interior optimum exists.

The empirical environment is described by award-record data: each firm i has a participation count tenders_count_i, a win indicator $\text{wins}_i \in \{0, 1\}$, and a tenure τ_i. The construct flags firms with $\text{wins}_i = 0$ and tenders_count_i above the threshold 14. Lemma 1 below formalizes the role of the wins-zero condition; Proposition 2 formalizes the role of the participation-count ranking.

Lemma 1 (Wins-zero separating equilibrium). *Suppose cover bidders draw their bid b_C from a strategy that satisfies (a) $b_C > b^*$ in equilibrium (cartel-allocation constraint) and (b) winning triggers either capacity revelation (undermining the cover identity in subsequent rounds) or internal cartel sanctions on the deviating firm. Genuine entrants of type G face no such constraint and accept any $b_G < b^*$ that yields positive expected payoff. Then in the separating equilibrium, cover bidders satisfy $\Pr(win \mid C) = 0$: the bright-line filter $wins = 0$ is the equilibrium choice of type C.*

Proof of Lemma 1. Write the cover-bidder per-period payoff as $u_C(b) = -c_1 - \theta_k\phi_0 \cdot \mathbf{1}[\text{win}] - \kappa \cdot \mathbf{1}[\text{win}]$, where $\phi_0 > 0$ is the marginal expected detection penalty per cover bid and $\kappa > 0$ captures expected cartel-internal sanction or capacity-revelation cost. Choosing $b_C > b^*$ delivers $\Pr(\text{win} \mid b_C, b^*) = 0$ in any equilibrium where the designated winner submits b^*, so $u_C = -c_1$. Any deviation to $b'_C < b^*$ delivers $u'_C = -c_1 - \theta_k\phi_0 - \kappa + \text{rents}$, where rents is the deviation gain from winning. Under the cartel's allocation rule, rents $<$ κ for any cover bidder (otherwise the allocation would not be incentive-compatible). Hence $u_C > u'_C$, and the equilibrium strategy is $b_C > b^*$ with $\Pr(\text{win} \mid C) = 0$. By contrast, type G has no κ penalty and accepts $b_G < b^*$ when expected payoff is positive. Hence the empirical filter wins $= 0$ identifies the type-C subset, conditional on the cartel deploying its allocation rule. □ □

Proposition 2 (Linearized cumulative exposure). *Suppose firm i is a cover bidder of type C deployed by a cartel in market k at per-period rate $\lambda_C(\theta_k, c_1)$ and tenure τ_i. Under A1–A2 and a stationary deployment process,*

$$\mathbb{E}[\, tenders\_count_i \mid type = C,\, \tau_i] \;=\; \lambda_C(\theta_k, c_1)\, \tau_i.$$

For genuine competitive entrants of type G, $\mathbb{E}[tenders\_count_i \mid type = G, \tau_i] = \lambda_G \tau_i$ with $\lambda_G < \lambda_C$ in any cartel-targeted market: by Proposition 3 below, $m^* > 0$ implies positive equilibrium deployment of cover bidders, while genuine entry does not respond to cartel demand for participation slots. The log-statistic $\log(1+tenders\_count_i)$ is therefore monotone in λ given tenure, and—under monotone likelihood ratio in the Poisson approximation $T_i \sim \text{Poisson}(\lambda\tau_i)$—the posterior $\Pr(type = C \mid tenders\_count_i = t)$ is non-decreasing in t. Combined with Lemma 1, the joint statistic $(\log(1 + tenders\_count), \mathbf{1}[wins = 0])$ is a sufficient screen for the cover-bidder type given award-record data; the binary frequent-loser rule is the information-coarsening of this latent posterior under the separating equilibrium.

Proof of Proposition 2. The expectation result is immediate from the law of large numbers applied to the deployment counting process given constant rate λ. The monotonicity claim follows from Bayes' rule:

$$\Pr(C \mid t) = \frac{\Pr(t \mid C)\, \pi_C}{\Pr(t \mid C)\, \pi_C + \Pr(t \mid G)\, (1 - \pi_C)},$$

which is non-decreasing in t if and only if the likelihood ratio $\Pr(t \mid C)/\Pr(t \mid G)$ is non-decreasing in t. For Poisson likelihoods with rates $\lambda_C > \lambda_G$, this monotone-likelihood-ratio property is standard. Hence $\log(1 + \text{tenders\_count})$ is a sufficient ranking statistic for the cover-bidder type, and the binary indicator $\mathbf{1}[\text{tenders\_count} > k]$ is its information-coarsening form. □ □

Proposition 3 (Optimal cover-bidder deployment). *Under A1–A2, the cartel's optimal cover-bidder deployment rate m^* satisfies the first-order condition $\partial\pi/\partial m = c_1 + \theta_k\phi_0$, where $\phi_0 > 0$ is the marginal expected detection penalty per cover bidder evaluated at the optimum. The implicit comparative statics are*

$$\frac{\partial m^*}{\partial \theta_k} = -\frac{\phi_0}{\partial^2\pi/\partial m^2} \ < \ 0, \qquad \frac{\partial m^*}{\partial c_1} = -\frac{1}{\partial^2\pi/\partial m^2} \ < \ 0.$$

Proof of Proposition 3. The first-order condition follows directly from differentiating $\pi(m, \theta_k, c_1)$ with respect to m and equating the marginal cartel surplus to the marginal cost of the cover bidder (c_1 deployment cost plus expected detection penalty $\theta_k\phi_0$). By A2, $\partial^2\pi/\partial m^2 < 0$, so the implicit interior solution $\tilde{m}$ defined by the FOC is unique. Implicit differentiation of the FOC yields the comparative statics displayed, both negative because the denominator is negative. The empirical implication is that cover-bidder deployment is more aggressive where the procuring unit's principal cost of detection is lower (§7.3). □ □

Why the screening object differs from a treatment-effect target. Propositions 2 and 3 together imply that frequent-loser participation is not distributed uniformly over items: under A1–A2, the cartel deploys cover bidders preferentially in items where its expected surplus is highest, conditional on the deployment cost c_1 and the local detection probability θ_k. Items the cartel selects into therefore differ from items it does not on a vector of unobserved characteristics $\mathbf{u}$ that governs the cartel's expected surplus—chiefly, the rents available from the designated winner's bid relative to the competitive equilibrium price, the local detection environment, and item-level characteristics that shape the cartel's allocation rule. None of these unobservables is captured by the fixed-effects controls or by the matching covariates: by construction, item, year, and procuring-unit dummies absorb sources of variation that are common to items within their respective strata, and the matching covariates are observable descriptors of the items themselves, while $\mathbf{u}$ is the unobservable component of the cartel's deployment optimum.

The following proposition formalizes the implication for the two empirical objects β and β^{ov}.

Proposition 4 (Screening-value vs. treatment-effect objects). *Let $D_i \in \{0,1\}$ indicate frequent-loser presence in item i, $\mathbf{x}_i$ the vector of observables absorbed by fixed effects and matching covariates, and $\mathbf{u}_i$ the vector of unobservables governing the cartel's deployment optimum (rents available, local detection environment, item-specific allocation features). Suppose the cartel's optimal deployment satisfies $\Pr(D_i = 1 \mid \mathbf{x}_i, \mathbf{u}_i) = g(\mathbf{x}_i, \mathbf{u}_i)$ with g strictly increasing in the rent component of $\mathbf{u}_i$. Then:*

1. Broad-sample β. *The conditional mean difference $\beta = \mathbb{E}[y_i \mid D_i = 1, \mathbf{x}_i] - \mathbb{E}[y_i \mid D_i = 0, \mathbf{x}_i]$ integrates over the distribution $f(\mathbf{u}_i \mid D_i = 1, \mathbf{x}_i)$ versus $f(\mathbf{u}_i \mid D_i = 0, \mathbf{x}_i)$. Under the cartel's deployment optimum, these two conditional distributions of $\mathbf{u}_i$ are systematically different in the rent component: items the cartel selects into have unobservable-shifted rents.*
2. Overlap-restricted β^{ov}. *The overlap-restricted comparison $\beta^{ov} = \mathbb{E}[y_i \mid D_i = 1, \mathbf{x}_i, \Omega] - \mathbb{E}[y_i \mid D_i = 0, \mathbf{x}_i, \Omega]$, where Ω denotes the common-support cell, restricts attention to items in which $D_i = 1$ and $D_i = 0$ have non-trivial joint density given $\mathbf{x}_i$. Within Ω, the conditional distributions of $\mathbf{u}_i$ are by construction more comparable: items where the cartel deploys at the $\mathbf{x}_i$-margin must compete with items where the cartel does not deploy despite similar observables, so the rent component of $\mathbf{u}_i$ that drives the deployment optimum must be smaller (otherwise the cartel would deploy with near-certainty there as well).*
3. Diagnostic. *If the rent component of $\mathbf{u}_i$ that drives the deployment optimum also moves prices in the same direction (rent-rich items both attract cartel deployment and realize higher prices conditional on observables), then $\beta > 0$ on the broad sample while $\beta^{ov} < 0$ is* the prediction *of the deployment problem, not its refutation: forcing overlap removes the cells where rents are high, which removes precisely the items where the participation footprint and the price imprint co-occur.*

Proof of Proposition 4. Claim 1 follows from the law of iterated expectations: the broad-sample β averages $\mathbb{E}[y_i \mid D_i, \mathbf{x}_i, \mathbf{u}_i]$ over the conditional distributions of $\mathbf{u}_i$ given D_i and $\mathbf{x}_i$. The cartel's deployment optimum implies that $\mathbf{u}_i \mid D_i = 1$ has a higher rent component than $\mathbf{u}_i \mid D_i = 0$ at the same $\mathbf{x}_i$, by Proposition 3. Claim 2 follows from the definition of common support: cells in Ω are those where the propensity score $\Pr(D_i = 1 \mid \mathbf{x}_i)$ is bounded away from 0 and 1, which by the strict monotonicity of g in the rent component implies that the rent component takes intermediate values. The $\mathbf{u}_i$-conditional distributions across $D_i = 1$ and $D_i = 0$ within Ω are therefore more comparable than across the broad sample. Claim 3 is the comparative statement: if the rent component shifts both the deployment probability and the realized price in the same direction, then averaging over $\mathbf{u}_i$ on the broad sample produces a positive β while restricting to Ω trims the high-rent cells out of the average, leaving a β^{ov} that need not be positive and may be negative for reasons unrelated to the treatment itself. □ ☐

The proposition formalizes the screening-value reading in §7.2. The broad-sample β is informative because it integrates over the distribution of items where the cartel deploys; under the screening framing of §5.1 this integration is the empirical object of interest. The overlap-restricted β^{ov} trims out exactly the cells where the deployment optimum places the cartel, which is what produces the sign reversal under Proposition 4's premise. The model does not by itself adjudicate whether the rent-component-co-movement premise holds; the empirical evidence of §6 (cartel-adjacency validation moves with β), §8.1 (continuous primitive dose-response), and §7.1 (sensitivity bounds) are the three independent observations that, together, make the screening-value reading more consistent with the data than a confound-only reading of β.

Appendix B. Core Supporting Tables and Figures

This appendix collects the supporting tables referenced throughout the body. Section Appendix B reports the sample-construction tables (modal mix, conditional descriptive statistics) and the heterogeneity tables (detection-regime quartiles, modal contrast, McCrary density test, robustness to exclusion of CADE-involved items). Figure B.1 displays the always-loser participation distribution that motivates the median+1.5×IQR cutoff; Figure B.2 shows the cutoff itself.

Appendix C. Identification Audits

The four audits referenced in §5.3 are reported here in detail. Table C.1 reports the Cinelli and Hazlett (2020) robustness value and the Oster (2019) bounds for the broad-sample association. Table C.2 reports the leakage decomposition that separates the structural component of the in-sample item-level discrimination AUC from the tautology component. Table C.3 reports the permutation-test design behind the conservative-benchmark co-bidding rate of §6.2: the participation-stratified co-bidding rate among

Table B.1: Tender Counts by Procurement Modality and Year

Year	Pregão	Convite	Total
2009	137,424	232,765	370,189
2010	134,181	253,196	387,377
2011	72,894	140,305	213,199
2012	129,664	305,808	435,472
2013	125,324	328,616	453,940
2014	118,274	310,947	429,221
2015	92,324	226,503	318,827
2016	57,804	122,841	180,645
2017	120,194	261,205	381,399
2018	60,518	121,698	182,216
2019	11,857	26,976	38,833

Notes: Winner-level observations from BEC (São Paulo) 2009–2019. Pregão = electronic reverse auction; Convite = sealed-bid invitation.

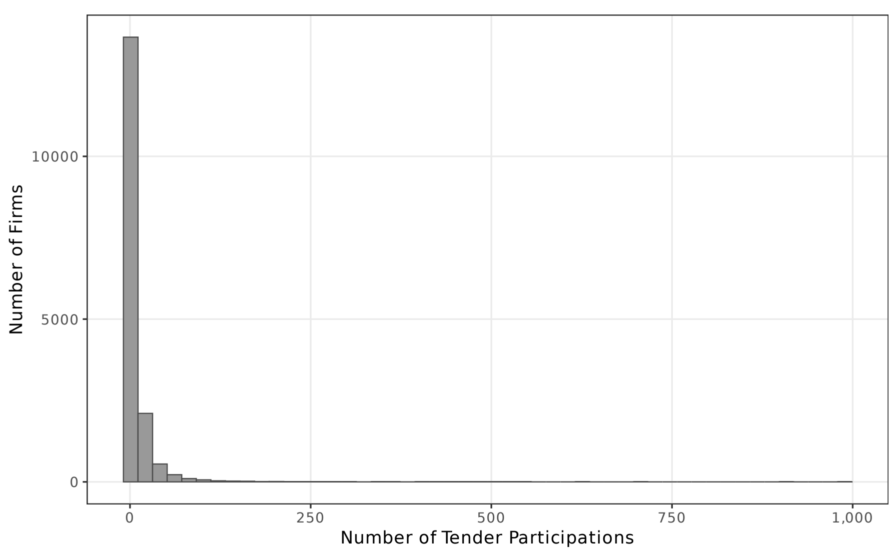

Figure B.1: Distribution of tender-item participation counts across always-loser firms in BEC, 2009–2019. The distribution is heavily right-skewed; a small share of always-losers participates in tens or hundreds of tender-items without ever winning.

Table B.2: Conditional Descriptive Statistics: Residualized Prices by FL Status

	FL-Present (losers = 1)	FL-Absent (losers = 0)
Panel A: Unconditional		
Mean log(price)	4.9241	2.4938
Difference	2.4303	
Panel B: Conditional on Item + Year + PBU FE		
Mean residualized price	0.0472	-0.0024
SD residualized price	1.5734	0.8704
Difference (gap)	0.0495	
Overlap coefficient	0.978	
t-statistic	8.81	
p-value	0.000000	
Cohen's d	0.0390	
Observations	79,452	1,574,949

Notes: Residuals from regressing log(negotiated price) on item, year, and purchasing unit (PBU) fixed effects. The gap represents the mean difference in residualized prices between FL-present and FL-absent tenders, after absorbing common item-level, temporal, and PBU-level variation. Overlap coefficient = $2\Phi(-|\Delta\mu|/\sqrt{\sigma_1^2 + \sigma_0^2})$ measures the proportion of the two residual distributions that overlap; values near 1 indicate near-identical distributions. Cohen's d measures the standardized effect size.

Table B.3: Frequent-Loser Price Coefficient by Procuring-Unit Size and Procedure

Sub-sample	Coefficient	SE	N
By procuring-unit size quartile:			
Q1 (smallest buyers)	0.2138	(0.1670)	7,503
Q2	0.0111	(0.0513)	87,571
Q3	0.0660**	(0.0304)	302,258
Q4 (largest buyers)	0.0165	(0.0160)	1,257,069
By procedure type:			
Pregão (reverse electronic auction)	0.0933***	(0.0255)	546,549
Convite (sealed-bid invitation)	0.0382**	(0.0186)	1,107,852

Notes: Quartile gradient consistent with variation in the screening signal across detection regimes (§7.3); we do not identify a causal channel. Item and year FE (PBU FE for procedure split). SE clustered at item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

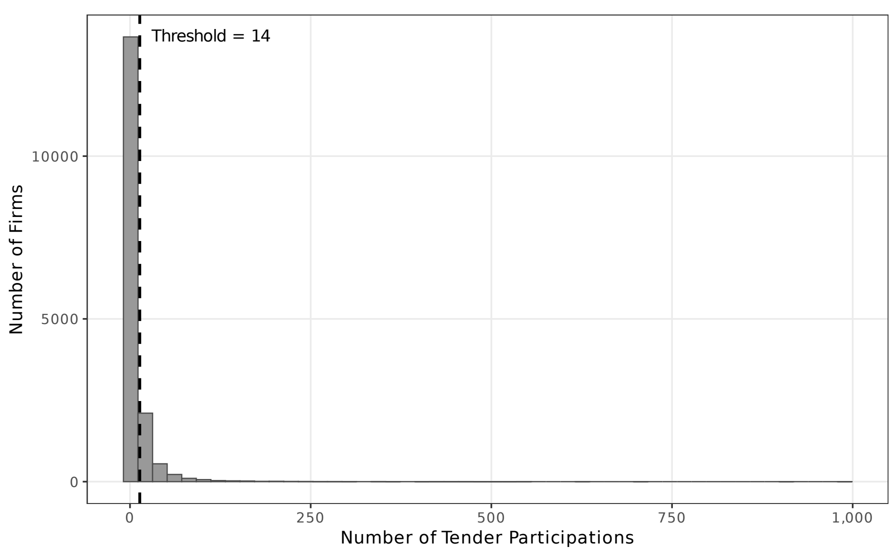

Figure B.2: Identification of the frequent-loser threshold via median + 1.5× IQR on the always-loser participation distribution. The cutoff yields 14 tender-items (13.5 statistical threshold) and flags 2,735 firms.

Table B.4: Frequent-Loser Price Coefficient by Modality (Pregão vs. Convite)

Specification	Pregão-only	Convite-only	Full sample	Ratio (P/C)
Binary FL14	$+0.0959^{***}$	$+0.0392^{**}$	$+0.0653^{***}$	2.45×
	(0.0256)	(0.0188)	(0.0216)	
	$p < 10^{-3}$	$p = 0.037$	$p = 0.003$	
Continuous log(1+ max tc)	$+0.0262^{***}$	$+0.0124^{**}$	—	2.11×
	(0.0063)	(0.0049)	—	
	$p < 10^{-4}$	$p = 0.011$	—	
Joint specification: both regressors				
Binary FL14 (with continuous control)	-0.0652	-0.0828^{**}	—	—
	(0.0449)	(0.0350)	—	—
Continuous (with binary control)	$+0.0399^{***}$	$+0.0305^{***}$	—	—
	(0.0118)	(0.0101)	—	—
Items	543,752	1,105,852	1,653,658	—
Item, year, PBU FE	YES	YES	YES	—
Cluster (item)	YES	YES	YES	—
Minimum-bidder rule binds	NO	YES	MIXED	—

Notes: Outcome is $\log p_{\text{negotiated}}$. The pregão coefficient is roughly 2.45× the convite coefficient on the binary specification. Under the screening framing (§5.1), we read the modal contrast as variation in the observability regime within which the screening signal is recovered, not as a positive test of any specific institutional channel. SEs clustered at item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.5: McCrary Density Diagnostic at the Convite/Pregão Statutory Threshold

	Density at threshold			Modality
Threshold (Lei 8.666 art. 23)	Left	Right	Log-density disc.	Convite share gap
R\$80,000 (pre-2018)	381	406	−0.063	0.098
R\$176,000 (post-2017)	216	277	−0.250	0.092

Notes: Densities estimated by local linear smoothing of contract-record counts in ±50% windows around each statutory cap. The convite-share gap is the difference between the share of contracts using the convite modality just below versus just above the threshold. Both thresholds show small log-density discontinuities ($|\cdot| < 0.25$); neither shows the kind of bunching that would indicate strategic gaming of the modality assignment by buyers. The ~ 9.8 percentage-point convite-share drop across the cap is consistent with rule-driven assignment of value to modality, not with strategic selection.

Table B.6: Price Regressions Excluding CADE-Involved Markets

	(1) General+PBU	(2) Pregão
FL presence	0.0616***	0.0901***
	(0.0213)	(0.0256)
Observations	1,622,954	534,956
CADE tenders dropped	12,514	

Notes: All tenders involving any CADE-convicted firm are excluded. SE clustered at item level. *** p<0.01, ** p<0.05, * p<0.1.

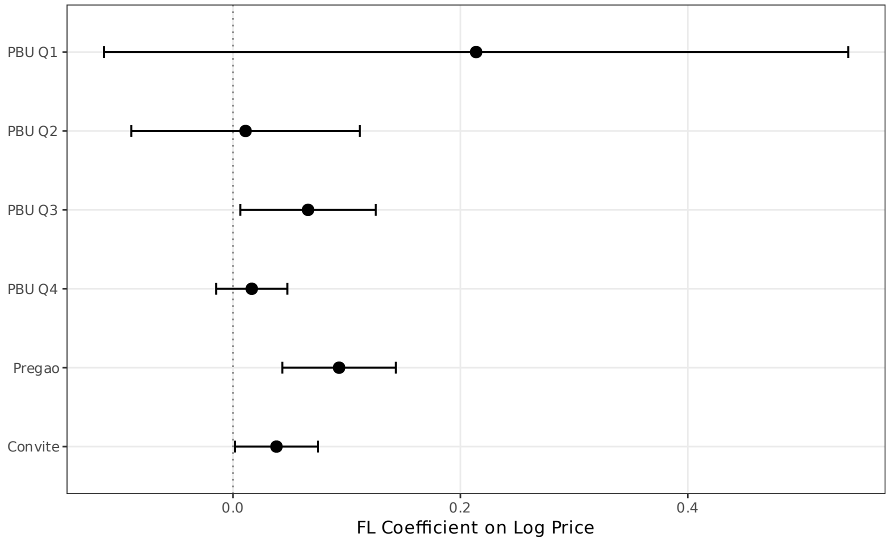

Figure B.3: Frequent-loser price coefficient by procuring-unit-size quartile. The Q1-vs-Q4 contrast (smallest to largest buyers) is 12.6×; the intermediate quartiles are imprecisely estimated. Companion figure to Table B.3; we read the contrast as detection-regime variation rather than as identification of a causal channel (§7.3).

frequent losers (3.95%) is contrasted with the empirical null distribution under random matching at comparable participation volume (1.24%, 1,000 iterations, $p < 0.001$).

Appendix D. Threshold and Specification Robustness

The IQR-multiplier sensitivity is reported in Table D.1: the headline price coefficient is positive and significant under five multipliers ranging from 1.0× to 3.0× IQR, declining monotonically as the cutoff tightens (the framework's continuous primitive predicts this pattern; §8.1). Standard-error clustering is reported in Table D.2 under item-level (baseline), procuring-unit-level, and two-way Cameron et al. (2011) clustering on item and PBU jointly. The coefficient is significant at $p < 0.01$ under all three regimes.

Appendix E. Generalization Audit and Detector Comparison

The temporal-holdout audit referenced in §9.3 is reported in Table E.1, and the same-sample incremental-AUC comparison referenced in §10 is reported in Table E.2. Table E.3 reports the cross-sector external-validity scope referenced in §11.3: the within-product-code identifying variation that underwrites the headline price gap is unevenly distributed across BEC item groups, and the construct's portability to non-commodity

Table C.1: Oster (2019) Coefficient Stability Bounds

	Full sample	Pregão	Convite
$\hat{\beta}_{\text{short}}$ (item + year FE)	0.0676	0.0954	0.0507
$\hat{\beta}_{\text{full}}$ (+ PBU FE)	0.0636	0.0933	0.0382
$\tilde{R}^2_{\text{short}}$	0.8790	0.8737	0.8857
$\tilde{R}^2_{\text{full}}$	0.8860	0.8821	0.8932
$R^2_{\text{max}} = \min(1, 1.3\tilde{R}^2)$	1.0000	1.0000	1.0000
$\hat{\delta}$ (Oster bound)	261.6	634.00	43.62
$\beta^*(\delta = 1)$	-0.0005	0.0643	-0.1395
Observations	1,654,401	546,549	1,107,852

Notes: $\hat{\delta}$ is the degree of selection on unobservables (relative to observables) required to drive $\hat{\beta}$ to zero (Oster, 2019). $|\hat{\delta}| > 1$ indicates robustness: unobservables would need to be more important than observables to explain the coefficient. $\beta^*(\delta = 1)$ is the bias-adjusted coefficient assuming equal selection. $R^2_{\text{max}} = 1.3\tilde{R}^2$ following Oster's recommended bound. Short model: item + year FE. Full model adds PBU FE (and convite control for full sample).

settings (services, infrastructure, public works) depends on the availability of analogous within-category variation in those settings.

Appendix F. Adversarial-Adaptation Simulation

The §11.2 discussion notes that the construct's behavior under deliberate adversarial adaptation is bounded but not invariant. This appendix reports the simulation behind that claim. Four strategic adaptations are examined; each holds the screening statistic fixed and modifies the always-loser pool to mimic plausible evasive behavior; the firm-level AUC of log(1+tenders_count) against the 193 CADE-cobidder labels is recomputed under each. Two are observable adaptations (rotation and occasional wins) that the construct is resilient to; two are threshold-aware adaptations (CNPJ splitting and participation capping) that degrade discrimination.

Resilient adaptations. Rotation (20% of cover bidders retire annually and are replaced by fresh CNPJs) and occasional wins (5% of threshold-crossing firms win one tender) leave

Table C.2: Leakage Audit of the Item-Level Loser-Side Concentration AUC

Audit specification	What it tests	AUC	95% CI
Original (in-sample, full pool)	Reference: log(1+max tc) predicting any-cobidder participation.	0.995	$[0.995, 0.995]$
Audit 1: Different label, same score (scope)			
Same score → any-direct-CADE label	Whether the screen also flags items with direct CADE defendants (N_+=54,039 items).	0.506	$[0.505, 0.507]$
Audit 2: 5-fold cross-validation at the cobidder-firm level (tautology)			
Pooled out-of-fold	Hold out $\sim 1/5$ of cobidders per fold; recompute tenders_count without their participation; predict items containing held-out cobidders using the recomputed score. Pool out-of-fold predictions across folds.	0.891	$[0.887, 0.894]$
Audit 3: Temporal holdout (generalization)			
Train 2009–2016 → test 2017–2019, any-cobidder	Compute tenders_count using only 2009–2016 items; predict items in 2017–2019.	0.864	$[0.859, 0.870]$
Train 2009–2016 → test 2017–2019, any-direct-CADE	Same temporal split, against direct-CADE label.	0.511	$[0.510, 0.513]$

Notes: The in-sample AUC of 0.995 is partly tautological by construction. Audit 2 isolates the structural component by holding cobidder firms out of score construction within each fold; Audit 3 isolates it temporally by computing the score on 2009–2016 participation only. AUC retains ≥ 0.85 under both audits; the 0.10–0.13 drop is the pure-leakage component. Audits against direct-CADE labels return random AUC, consistent with the scope asymmetry in Table 5.

Table C.3: CADE Validation: Co-Participation and Permutation Test

Panel A: Co-participation with CADE Cartelists		
CADE procurement cartel cases (unique processes)	12	
firm-defendants across all cases	65	
final adjudication dates	2015–2025	
cases adjudicated after 2019 (prospective)	8	
CADE firm-defendants matched to BEC via CNPJ	47	
	FL Firms	Non-FL Firms
Co-bids with CADE	193	2,290
Does not co-bid	2,542	36,419
Total	2,735	38,709
Co-participation rate	7.1%	5.9%
χ^2 statistic	5.7	
p-value	0.017	
Odds ratio	1.21	
Panel B: Permutation Test (1,000 iterations)		
Observed FL-CADE rate	0.0706	
Permutation mean rate	0.0201	
Permutation SD	0.0026	
p-value (rate $\geq$ observed)	0.0000	

Notes: Panel A: contingency table of firm co-participation with CADE cartel-defendants. Eight of the 12 cases were adjudicated after the BEC sample ends (post-2019), so the validation is partially prospective: it asks whether 2009–2019 FL flags anticipate firms later confirmed by enforcement. Panel B: 1,000 random draws of 2,735 firms from the always-loser pool, stratified by participation quartile, computing co-participation rate with CADE firm-defendants.

Table D.1: Price Coefficient Across IQR Multiplier Thresholds

Multiplier	Threshold	FL Firms	Coef.	SE	Observations
1.0 × IQR	10	3,442	0.0791***	(0.0227)	1,654,401
1.5 × IQR	14	2,735	0.0636***	(0.0215)	1,654,401
2.0 × IQR	17	2,093	0.0598***	(0.0221)	1,654,401
2.5 × IQR	20	1,778	0.0543**	(0.0228)	1,654,401
3.0 × IQR	24	1,456	0.0501**	(0.0237)	1,654,401

Notes: DV: log negotiated price. Each row re-estimates the baseline specification (item + year + PBU fixed effects) using a different IQR multiplier to define the FL threshold. The threshold equals median $+k\times$ IQR of the tenders-count distribution among always-losers. The baseline (1.5× IQR) is highlighted. SE clustered at item level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table D.2: Alternative Clustering of Standard Errors

Clustering level	Coef.	SE	95% CI
Item level (baseline)	0.0636***	(0.0215)	[0.0214, 0.1058]
PBU level	0.0636***	(0.0144)	[0.0353, 0.0919]
Two-way (item + PBU)	0.0636***	(0.0240)	[0.0165, 0.1107]

Notes: DV: log negotiated price. All rows report the same baseline specification (item + year + PBU fixed effects) with different clustering of standard errors. The point estimate is invariant to clustering; only the standard errors and confidence intervals change. The two-way cluster uses the Cameron et al. (2011) multi-way approach. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table E.1: Operational Deployment Metrics: Precision, Recall, and Lift at Top-k

	In-sample (full 2009–2019)				Temporal holdout (score: 2009–2016)			
Top-k	TP	Precision	Recall	Lift	TP	Precision	Recall	Lift
50	15	0.300	0.078	26.2×	1	0.020	0.005	1.7×
100	17	0.170	0.088	14.8×	7	0.070	0.036	6.1×
250	40	0.160	0.207	14.0×	19	0.076	0.098	6.6×
500	66	0.132	0.342	11.5×	35	0.070	0.181	6.1×
1,000	97	0.097	0.503	8.5×	66	0.066	0.342	5.8×
Reference quantities								
Always-loser pool size N					16,843			
Cobidder positives $N+$					193			
Base cobidder rate					0.0115			
ROC AUC (log(1+tc))	In-sample: 0.939				Temporal holdout: 0.864			

Notes: Score: log(1 + tenders_count), ranked descending across the always-loser pool. Lift is precision divided by the base cobidder rate. The in-sample columns score firms using 2009–2019 participation counts; the temporal-holdout columns score using only 2009–2016 participation, evaluated against the same 193 cobidder labels. The temporal-holdout numbers are the generalization reference for the screening statistic; the in-sample columns over-state precision by approximately 50% at top-500 and are reported for transparency. The screening statistic requires only contract-award records (winner identity, participant identity, item identifier); no bid microdata.

Table E.2: Incremental Value of Loser-Side Concentration Relative to the Full Imhof Benchmark

Model	Data requirements	AUC	95% CI	Δ vs Imhof full	DeLong p
Imhof full pipeline	All bid values within each tender	0.846	$[0.819, 0.873]$	+0.000	—
Binary FL flag	Winner + participants only	0.881	$[0.871, 0.892]$	+0.035	0.014
Continuous participation count	Winner + participants only	0.877	$[0.857, 0.898]$	+0.031	0.077
Imhof full + binary FL	Bid microdata + award records	0.942	$[0.927, 0.957]$	+0.096	< 0.001
Imhof full + participation count	Bid microdata + award records	0.944	$[0.929, 0.958]$	+0.098	< 0.001

Notes: All models are evaluated on the exact same always-loser subsample for which the full bid-distribution feature set is available. The relevant comparison is therefore not "does FL beat Imhof everywhere," but whether participation-only information remains informative when rich bid microdata are absent and whether it adds non-redundant signal when those microdata are present. The combined specifications answer the second question directly.

Table E.3: External Validity Scope: Modality and Commodity-Service Boundaries

Dimension	Group	Metric	Value	95% CI
coverage	Commodity	share_items	0.887	—
coverage	Service	share_items	0.113	—
dominant_item_class	Commodity	auc_log_tc	0.905	[0.879, 0.931]
dominant_item_class	Service	auc_log_tc	0.966	[0.961, 0.971]
item_class_price	Commodity	coef_losers	0.045	[0.002, 0.088]
item_class_price	Service	coef_losers	0.128	[0.024, 0.231]
modal_primary_auc	convite_primary	auc_log_tc	0.816	[0.758, 0.874]
modal_primary_auc	pregao_primary	auc_log_tc	0.952	[0.946, 0.958]

Notes: Item classes use a transparent text rule on BEC item descriptions. Rows labelled `modal_primary_auc` reuse the always-loser firm split by primary modality; rows labelled `item_class_price` report the within-item price association separately for commodity-like and service-like items. The point is not to claim exportability to all procurement, but to document that the paper speaks most naturally to standardized-goods environments and selected simple services. *Source:* `scripts/52_external_validity_scope.R`.

Table F.1: Adversarial-Adaptation Simulation: AUC Degradation Under Strategic Cartel Adaptations

Scenario	AUC	Δ vs baseline	Retention
Baseline (no adaptation)	0.928	+0.000	100.0%
Rotation: 20% of cover bidders retire and are replaced	0.931	+0.003	100.3%
Occasional wins: 5% of FL14 firms get one win	0.929	+0.002	100.2%
CNPJ splitting: each FL14 firm splits into two CNPJs	0.891	−0.037	96.0%
Threshold cap: firms cap participation at 13 tender-items	0.801	−0.127	86.3%
Combined: threshold cap + 20% rotation	0.632	−0.295	68.2%

Notes: Each scenario applies a strategic adaptation to the BEC always-loser pool, holding the construct (log(1+tenders_count) ranking against 193 CADE cobidder labels) fixed. The construct is resilient to rotation and to occasional wins (the bright-line filter eliminates the small share of newly-winning cover bidders without compressing the ranking), and degrades modestly under CNPJ splitting. It is substantially vulnerable to threshold-aware capping and severely degraded when capping is combined with rotation: a sophisticated cartel that targets the threshold can compress AUC by roughly 30 points. Retention is the fraction of baseline AUC preserved.

AUC essentially unchanged. The bright-line wins $= 0$ filter eliminates the small share of newly-winning cover bidders—a feature of the framework's separating equilibrium (Lemma 1), not a heuristic robustness—and rotation does not compress the $\log(1 + \text{tenders\_count})$ ranking statistic across the always-loser pool. The screening statistic is structurally robust to adaptations that preserve the participation primitive's distribution.

Threshold-aware adaptations. CNPJ splitting (each frequent loser splits into two CNPJs carrying half the participation count) degrades AUC by 3.7 percentage points; threshold-aware capping (firms cap participation at 13 tender-items to stay below the cutoff) degrades AUC by 12.7 points; the combined attack (capping plus 20% rotation) drives AUC from 0.928 to 0.632, a 29.5-point compression. These adaptations target the binary cutoff specifically; the continuous primitive $\log(1 + \text{tenders\_count})$ that Proposition 2 identifies as the sufficient ranking statistic is not directly altered, but the binary discretization is. Threshold capping is therefore the most exposed margin: a cartel that knows the cutoff can instruct cover bidders to remain below it.

Architectural implication. The screening statistic is informative absent adaptation and degrades predictably under specified attacks. The implication for the screening-vs-forensic-stage architecture of §10 is asymmetric. Under combined adaptation, AUC drops to 0.632—still above random matching, still informative as a screening statistic, but substantially reduced. The forensic stage is unaffected by threshold-aware adaptation at the screening layer: bid-distribution moments operate on a different data envelope and are robust to the participation-pattern engineering that defeats the binary rule. A combined two-stage architecture in which a refreshable screening rule feeds a forensic stage on bid microdata is therefore more adaptation-robust than either component alone, which the empirical evidence in §10 shows by distinct means.

The screening statistic is not adaptation-proof and we do not claim it is. Practical lifetime depends on whether adaptation is observable to the enforcer (the threshold rule's exposed margin is itself observable, since the cutoff is published) and on whether the threshold can be periodically refreshed against new data. The honest characterization is that the construct is informative against an unsophisticated adversary, vulnerable to a sophisticated one targeting the binary rule, and best deployed alongside the bid-layer forensic stage where microdata are available—which is the architectural implication the body already states.

Appendix G. Staggered Designs Attempted and Rejected

The §11.1 discussion notes that two design-based strategies that would extract a causal estimate from the data return null at the available bandwidths. The sharp-RDD failure (gated by the McCrary density test) is documented in Table B.5 of App B. This appendix reports the parallel diagnostic for the difference-in-differences strategy exploiting Decreto 9.412/2018's 2018 raise of the convite cap. We report the failure transparently because the empirical content of the paper is descriptive throughout, and the screening framing of §5.1 does not require these designs to succeed; the documented null is part of the disclosure that locates the construct as a screening-value object rather than a causal-effect estimate.

The Sun and Abraham (2021) interaction-weighted estimator yields imprecise coefficients with non-monotone dynamic patterns incompatible with the smooth treatment-onset narrative the cap raise would imply (Table G.1). A stacked difference-in-differences design constructed around the cap-raise event window returns coefficients that flip sign across stacks and lack a clean pre-period (Table G.2). We treat both estimators as null at the available bandwidths.

The interpretive consequences are limited to identification scope. The screening statistic does not require the cap raise to be a clean treatment, the broad-sample association β of §7.1 does not depend on the staggered design surviving, and the null does not constrain the screening-value interpretation of §5.1. What the null does constrain is alternative framings of the empirical object that would require a causal identification of β; under those framings, the design-based strategies returning null limit the scope of any treatment-effect claim. The screening framing trades that claim for the screening-value interpretation of β and reports the failures here for full disclosure.

Table G.1: Staggered DiD: Callaway & Sant'Anna (Restricted Sample)

	C&S (2021)		TWFE	
Outcome	ATT	SE	Pre-trend	Post
Log Price	0.0143	(0.0396)	-0.0131	-0.0079
Log Firms (excl. FL)	-0.0220	(0.0333)	-0.0088	0.0113
Panel B: Statistical Power				
MDE (5%, two-sided)	0.0776			
MDE (80% power)	0.1109			
OLS benchmark	0.064			
Panel D: Sharp Entry Subsample				
C&S ATT	0.0143	(0.0389)		
Sharp entry markets	1,511 treated, 18,266 control			
Markets	19,777 (1,511 treated, 18,266 control)			
Sample restriction	$\geq$3 pre & $\geq$3 post observations			

Notes: Panel A: C&S = Callaway & Sant'Anna (2021) doubly-robust estimator. Panel B: MDE = minimum detectable effect at stated significance/power levels. Panel C: Rambachan & Roth (2023) sensitivity bounds under smoothness restriction $\bar{M}$ on maximum change in the slope of the trend. Panel D: restricts treated markets to those with zero FL presence before treatment year.

Table G.2: Stacked DiD vs. Callaway & Sant'Anna

	Stacked (Cengiz et al.)		C&S (2021)	
Outcome	ATT	SE	ATT	SE
Log Price	-0.0059	(0.0138)	0.0143	(0.0405)
Log Firms (excl. FL)	0.0162*	(0.0084)	-0.0220	(0.0339)
Stacked obs.		715,116		
Cohorts		6		
Event window		±5 years		
Cohort × Market FE	YES		—	
Cohort × Year FE	YES		—	
Clustering		Market level		

Notes: Stacked regression following Cengiz et al. (2019). Each cohort-year sub-experiment includes treated markets entering in year g and all never-treated markets, within a symmetric ±5-year event window. Cohort-specific market and year fixed effects absorb level differences across sub-experiments. SE clustered at market level. C&S = Callaway & Sant'Anna (2021) doubly-robust estimator from Table G.1. *** p<0.01, ** p<0.05, * p<0.1.

References

Cameron, A. C., J. B. Gelbach, and D. L. Miller (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics 29*(2), 238–249.

Cengiz, D., A. Dube, A. Lindner, and B. Zipperer (2019). The effect of minimum wages on low-wage jobs. *Quarterly Journal of Economics 134*(3), 1405–1454.

Cinelli, C. and C. Hazlett (2020). Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B 82*(1), 39–67.

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics 37*(2), 187–204.

Sun, L. and S. Abraham (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics 225*(2), 175–199.